

Equation Heavy Fixture

Strata Reader Project

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1 Introduction

This document contains over 30 mathematical equations in display mode to verify formula detection.

2 Mathematical Equations

Equation 1: Euler's Identity

$$e^{i\pi} + 1 = 0 \tag{1}$$

Equation 2: Einstein's Field Equations

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{2}$$

Equation 3: Schrödinger Equation

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \hat{H} \Psi(\mathbf{r}, t) \tag{3}$$

Equation 4: Maxwell's Equations - Gauss's Law

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \tag{4}$$

Equation 5: Maxwell's Equations - Gauss's Law for Magnetism

$$\nabla \cdot \mathbf{B} = 0 \tag{5}$$

Equation 6: Maxwell's Equations - Faraday's Law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{6}$$

Equation 7: Maxwell's Equations - Ampere's Law

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \tag{7}$$

Equation 8: Fourier Transform

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx \quad (8)$$

Equation 9: Inverse Fourier Transform

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x \xi} d\xi \quad (9)$$

Equation 10: Gaussian Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad (10)$$

Equation 11: Navier-Stokes Equation

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f} \quad (11)$$

Equation 12: Heat Equation

$$\frac{\partial u}{\partial t} - \alpha \nabla^2 u = 0 \quad (12)$$

Equation 13: Wave Equation

$$\frac{\partial^2 u}{\partial t^2} - c^2 \nabla^2 u = 0 \quad (13)$$

Equation 14: Black-Scholes Equation

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0 \quad (14)$$

Equation 15: Logistic Map

$$x_{n+1} = rx_n(1 - x_n) \quad (15)$$

Equation 16: Shannon Entropy

$$H(X) = - \sum_{i=1}^n P(x_i) \log_2 P(x_i) \quad (16)$$

Equation 17: Partition Function

$$Z = \sum_s e^{-\beta E_s} \quad (17)$$

Equation 18: Boltzmann Distribution

$$P(s) = \frac{1}{Z} e^{-\beta E_s} \quad (18)$$

Equation 19: Ideal Gas Law (Statistical Mechanics)

$$PV = Nk_B T \quad (19)$$

Equation 20: Planck's Law

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{\frac{h\nu}{k_B T}} - 1} \quad (20)$$

Equation 21: De Broglie Relation

$$\lambda = \frac{h}{p} \quad (21)$$

Equation 22: Dirac Equation

$$(i\gamma^\mu \partial_\mu - m) \psi = 0 \quad (22)$$

Equation 23: Bragg's Law

$$2d \sin \theta = n\lambda \quad (23)$$

Equation 24: Friedmann Equation

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{kc^2}{a^2} = \frac{8\pi G\rho + \Lambda c^2}{3} \quad (24)$$

Equation 25: Euler-Lagrange Equation

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = 0 \quad (25)$$

Equation 26: Hamilton's Equations

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i} \quad (26)$$

Equation 27: Central Limit Theorem (Characteristic Function)

$$\varphi_{\Sigma_n}(t) = \left[\varphi_X \left(\frac{t}{\sqrt{n}} \right) \right]^n \rightarrow e^{-\frac{t^2}{2}} \quad (27)$$

Equation 28: Laplace Transform

$$F(s) = \int_0^\infty e^{-st} f(t) dt \quad (28)$$

Equation 29: Stirling's Approximation

$$\ln n! \approx n \ln n - n \quad (29)$$

Equation 30: Riemann Zeta Function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} \quad (30)$$

Equation 31: Cauchy-Schwarz Inequality

$$|\langle \mathbf{u}, \mathbf{v} \rangle|^2 \leq \langle \mathbf{u}, \mathbf{u} \rangle \langle \mathbf{v}, \mathbf{v} \rangle \quad (31)$$

Equation 32: Taylor Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n \quad (32)$$